

Homeworks

1. Design ER model for conference management system that manages authors, papers, reviewers, sessions, and scheduling. Also write SQL DDL to convert this ER model to equivalent relational schemas.
2. Given relational schemas

Employee(eid, emp_name, address, supervisor_id)

Department(did, dep_name)

Project(pid, prj_name, did)

Workson(eid, pid, hours)

Write relational algebraic expression and SQL for performing following operations.

- i. List the name of all employee for computer department along with their supervisor Name.
- ii. Find the total number of projects from each department along with department name.
- iii. Find the employee's name who do not work in any project.
- iv. Find the name of each project with more than 3 employees.

Homework

3. Define stored procedure. How can we create parameterized stored procedure? Explain with necessary syntax and example.
4. Define Trigger and its role in maintaining data integrity. In a conference Management System that manages authors, papers, reviewers, sessions and scheduling, how can trigger be utilized to enforce constraint “A reviewer can review a maximum of 5 papers”.
5. Write the algorithm for insertion in B+ Index Tree. Construct B+ tree of order 4 for the following set of keys, 2,3,5,7,11,17,19,23,29,31.
6. In a university database, how can you efficiently get the total salary paid to instructor in each department without recalculating the sum every time? Write necessary SQL queries to create, refresh and view for this purpose.

Homework

7. Explain the difference between strong consistency, eventual consistency and casual consistency with suitable examples.
8. Can a System be strongly Consistent and highly available at the same time? Relate your answers to the CAP Theorem.

Homework

9. What is Sharding? Explain Sharding Operation and its implementation using any NoSQL database tools. [SOE Pre-board, 2025]
10. Write the Key features of NoSQL System. Describe how CRUD operations are performed using any NoSQL database tool of your choice. [PU, 2025]
11. Define SQL and NoSQL. Explain different types of NoSQL database with their data model, characteristics, advantages and disadvantages. [PU, 2025]

Homework

12. Explain the working principles of the MapReduce Programming Model. Describe the roles of Map and Reduce functions in distributed data processing, and illustrate your explanations with a real world use case. [PU, 2025]
13. Explain the working mechanism of blockchain technology with appropriate illustrations. Why is a blockchain based database considered more secure than traditional centralized database systems? [PU, 2025]
14. Explain the architecture of HDFS. How is read and write performed in HDFS? [SOE, Pre-board]

Homework

15. Differentiate between Redis and Memcached Caching Strategies. What are the various techniques for horizontal partitioning in databases? Explain any one of these techniques in detail. [PU, 2025]
16. Write short notes on: [PU, SOE]
 - a. Local and Global Indexing
 - b. Blockchain database

Homework

17. Write short notes on: [PU, SOE]

- a. Vector Database
- b. Edge Database
- c. Cloud Database